

GOSFORD HIGH SCHOOL.
Extension 2 Mathematics.
Assessment task December 2004.

Part A.

Question 1.

If $z = 3 + 4i$ and $\omega = 1 + i$ find in the form $a + ib$

- a) $z + \omega$ 1
- b) $z\omega$ 2
- c) $\frac{z}{\omega}$ 2
- d) $\bar{z}$ 1
- e) $|z|$ 2

Question 2.

Simplify $\frac{(\cos \theta + i \sin \theta)^9 \cdot (\cos 3\theta + i \sin 3\theta)^{-5}}{(\cos 2\theta - i \sin 2\theta)^4}$ 2

Question 3.

If $z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$ evaluate z^6 2

Question 4.

Factorise $z^2 + 4z + 5$ over the complex field. 2

Question 5.

- a) Find the square root of $-8 - 8i\sqrt{3}$ 3
- b) Hence solve the quadratic equation $x^2 - i2\sqrt{2}x + i2\sqrt{3} = 0$. 2

Question 6.

Given that $1, \omega, \omega^2$ are the cube roots of unity show that $\frac{1}{1+\omega} + \frac{1}{1+\omega^2} = 1$ 2

Question 7.

Write down the conjugate of $a+ib$ and hence show that if $z = x + iy$ 2

then $\frac{z + \bar{z}}{z\bar{z}}$ is real.

Question 8.

Find the modulus and argument of the quotient when $\sqrt{3} - i$ is divided by $-1 - i$. 3

Question 9.

Find all the fourth roots of 16 and show that if β_2 and β_4 are the two imaginary roots then $\beta_2^3 + 2\beta_2^2 + 4\beta_2 + 8 = 0$. 4

Question 10.

Find all values of z such that $z^5 + 1 = 0$. 2

Question 11.

Find $\cos 4\theta$ in terms of:

a) $\sin \theta$ and $\cos \theta$. 3

b) $\cos \theta$ alone. 2

c) Hence solve $8\cos^4 x - 8\cos^2 x + 1 = 0$ for $0 \leq x \leq \pi$. 3

Question 12.

Draw a neat sketch of the following.

a) $\text{Im}(z) = 4$ 1

b) $|z| \leq 4$ 2

c) $|z - 3 + 4i| = 5$ 2

d) $\arg(z + i) = \frac{\pi}{4}$ 2

e) $z\bar{z} - 4(z + \bar{z}) = 10$ 3

Question 13.

The quadratic equation $z^2 + (1 + i)z + k = 0$ has a root of $1 - 2i$. Find, in the form $a + ib$, the value of k and the other root of the equation. 3

Question 14.

Given $z = \cos \theta + i \sin \theta$

a) Show $z^n + \frac{1}{z^n} = 2 \cos n\theta$. 2

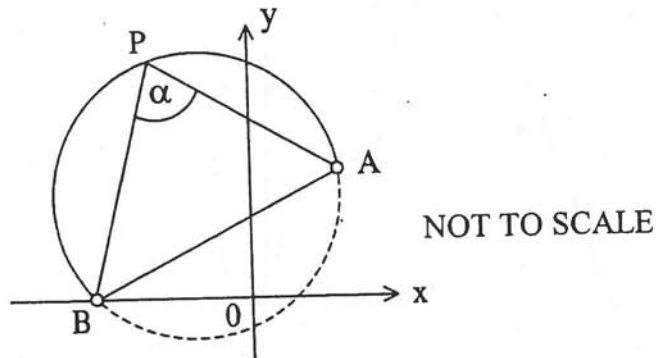
b) Hence using (a) and expanding $(z + \frac{1}{z})^3$ show that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta. \quad 3$$

Question 15.

What is the maximum value of $|z|$ for $|z - 1 - i| \leq 2$? 3

Question 16.



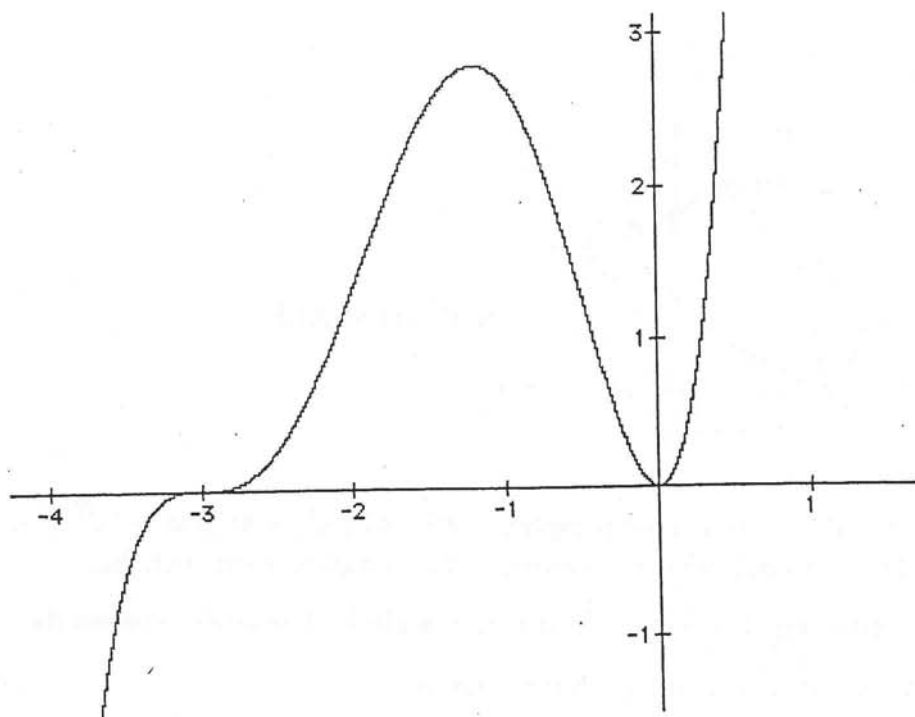
The points P, A and B represent the complex numbers Z , $\sqrt{3} + 4i$ and $-3\sqrt{3}$ respectively. The locus of Z which is moving in the complex plane such that $\text{Arg}(Z - \sqrt{3} - 4i) - \text{Arg}(Z + 3\sqrt{3}) = \frac{\pi}{3}$, is part of a circle. The angle between the lines from PA and PB is α as shown in the diagram.

- Show that $\alpha = \frac{\pi}{3}$. 2
- Find the centre and radius of the circle. 3

Part B continued over the page

Part B.

Question 1



- (a) Consider the graph of $y = f(x)$ as shown above.
On the answer sheet provided, use the graphs of $y = f(x)$ to clearly sketch separately the graphs of:
- | | |
|--------------------------|---|
| (i) $y = \frac{1}{f(x)}$ | 2 |
| (ii) $y^2 = f(x)$ | 2 |
| (iii) $y = f'(x)$ | 1 |
- (b) Suggest a possible polynomial equation for the graph of $y = f(x)$ shown in part (a)
- | | |
|--|---|
| | 1 |
|--|---|

Question 2.

- (a) If $f(x) = (x-1)(x-3)$ then sketch
- | | |
|-----------------------|---|
| (i) $y = \sqrt{f(x)}$ | 2 |
| (ii) $y = f(x)$ | 2 |
| (iii) $ y = f(x)$ | 2 |

(b) (i) Find the stationary points and the asymptotes of the function

2

$$y = \frac{(x+1)^4}{x^4+1}$$

(ii) Sketch this function labelling all essential features.

1

(iii) Use the graph to find the set of values of k for which

2

$(x+1)^4 = k(x^4+1)$ has two distinct real roots.

Question 3.

A curve is given parametrically in terms of the real number t by the equations

$$x = \frac{3t}{1+t^3} \quad \text{and} \quad y = \frac{3t^2}{1+t^3}.$$

(i) Express t in terms of x and y . Hence show that the curve has Cartesian equation

3

$x^3 + y^3 = 3xy$. Deduce that the curve is symmetrical about the line $y = x$.

(ii) Show that $\frac{dy}{dx} = \frac{y-x^2}{y^2-x}$. Hence show that the curve has a horizontal tangent when $x = \sqrt[3]{2}$.

3

Write down the coordinates of a point on the curve where the tangent is vertical.

PART A

Ques 1

$$z = 3 + 4i, \quad w = 1 + i$$

a) $z + w$

$$= 3 + 4i + 1 + i$$

$$= 4 + 5i$$

b) zw

$$= (3 + 4i)(1 + i)$$

$$= 3 + 3i + 4i - 4$$

$$= -1 + 7i$$

c) $\frac{z}{w}$

$$= \frac{3 + 4i}{1 + i} \times \frac{1 - i}{1 - i}$$

$$= \frac{(3 + 4i)(1 - i)}{1^2 - (i)^2}$$

$$= \frac{7 + i}{2}$$

$$= \frac{7}{2} + \frac{i}{2}$$

d) $\bar{z} = 3 - 4i$

e) $|z| = \sqrt{3^2 + 4^2}$
 $= 5$

Ques 2

$$\frac{(\cos \theta + i \sin \theta)^9 (\cos 3\theta + i \sin 3\theta)^5}{(\cos 2\theta - i \sin 2\theta)^4}$$

$$= \frac{(\cos \theta + i \sin \theta)^9 ((\cos \theta + i \sin \theta)^3)^5}{((\cos \theta + i \sin \theta)^{-2})^4}$$

$$\frac{(\cos \theta + i \sin \theta)^9 (\cos \theta + i \sin \theta)^{-15}}{(\cos \theta + i \sin \theta)^{-8}}$$

$$= (\cos \theta + i \sin \theta)^2$$

$$= \cos 2\theta + i \sin 2\theta$$

Ques 3

$$z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$$

$$z^6 = 2^6 (\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})^6$$

$$= 64 (\cos \pi + i \sin \pi)$$

$$= 64 (-1 + 0i)$$

$$= -64$$

Ques 4

$$z^2 + 4z + 5$$

$$= z^2 + 4z + 4 + 1$$

$$= (z + 2)^2 - i^2$$

$$= (z + 2 + i)(z + 2 - i)$$

Ques 5

a) let $a + ib = \sqrt{-8 - 8i\sqrt{3}}$

$$\therefore a^2 - b^2 + 2iab = -8 - 8i\sqrt{3}$$

equating real and imaginary parts

$$a^2 - b^2 = -8 \quad \dots (1)$$

$$2ab = -8\sqrt{3}$$

$$\therefore ab = -4\sqrt{3} \quad \dots (2)$$

$$(2) \Rightarrow b = \frac{-4\sqrt{3}}{a}$$

Sub into (1)

$$a^2 - \left(\frac{-4\sqrt{3}}{a}\right)^2 = -8$$

$$a^2 - \frac{48}{a^2} = -8$$

$$a^4 - 48 = -8a^2$$

$$a^4 + 8a^2 - 48 = 0$$

$$(a^2 - 4)(a^2 + 12) = 0$$

$$a^2 = 4 \quad \text{or} \quad a^2 = -12 \rightarrow \text{no solution}$$

$$\therefore a = \pm 2.$$

Sub into (2)

$$\therefore b = -2\sqrt{3}, 2\sqrt{3}.$$

$$\therefore \sqrt{-8-8i\sqrt{3}} = 2-3\sqrt{2}i, -2+2\sqrt{3}i$$

$$b) \quad x^2 - i\sqrt{2}x + i2\sqrt{3} = 0$$

$$x = \frac{i\sqrt{2} \pm \sqrt{-8-8i\sqrt{3}}}{2}$$

$$= \frac{2\sqrt{2}i \pm (2-2\sqrt{3}i)}{2}$$

$$= i\sqrt{2} \pm (1-\sqrt{3}i)$$

$$= 1+i(\sqrt{2}-\sqrt{3}), -1+i(\sqrt{3}-\sqrt{2})$$

$$Q6) \quad \frac{1}{1+\omega} + \frac{1}{1+\omega^2} = 1$$

L.H.S. (using $1+\omega+\omega^2=0$)

$$= \frac{1}{-\omega^2} + \frac{1}{-\omega}$$

$$= \frac{1+\omega}{-\omega^2}$$

$$= \frac{-\omega^2}{-\omega^2}$$

$$= 1$$

$$= \text{R.H.S.}$$

(2)

Ques 7

Conjugate $a+ib = a-ib$

$$\frac{z + \bar{z}}{z\bar{z}}$$

$$\frac{z\bar{z}}{z\bar{z}}$$

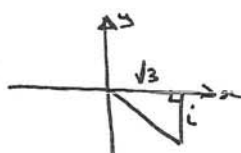
$$= \frac{x+iy + x-iy}{(x+iy)(x-iy)}$$

$$= \frac{2x}{x^2+y^2}$$

which is real.

Ques 8

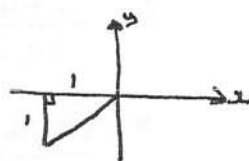
$$\sqrt{3}-i$$



$$|\sqrt{3}-i| = 2.$$

$$\arg(\sqrt{3}-i) = -\frac{\pi}{6}$$

$$-1-i$$



$$|-1-i| = \sqrt{2}$$

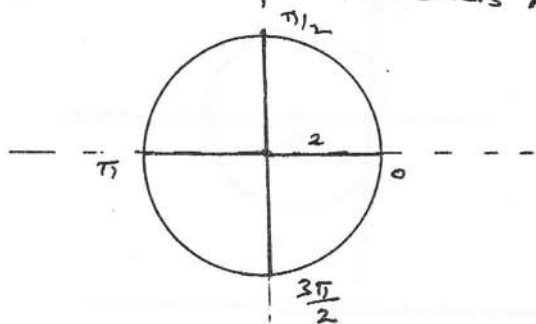
$$\arg(-1-i) = -\frac{3\pi}{4}$$

$$\therefore \left| \frac{\sqrt{3}-i}{-1-i} \right| = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\begin{aligned} \arg\left(\frac{\sqrt{3}-i}{-1-i}\right) &= \arg(\sqrt{3}-i) - \arg(-1-i) \\ &= -\frac{\pi}{6} - \left(-\frac{3\pi}{4}\right) \\ &= \frac{7\pi}{12}. \end{aligned}$$

Ques 9

roots are equally spaced around a circle, radius 2.



$$z_1 = 2(\cos 0^\circ + i \sin 0^\circ) = 2$$

$$z_2 = 2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = 2i$$

$$z_3 = 2(\cos \pi + i \sin \pi) = -2$$

$$z_4 = 2(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}) = -2i$$

Let $\beta_2 = 2i$, $\beta_4 = -2i$

$$\beta_2^3 + 2\beta_2^2 + 4\beta_2 + 8 = 0$$

L.H.S

$$= (2i)^3 + 2(2i)^2 + 4(2i) + 8$$

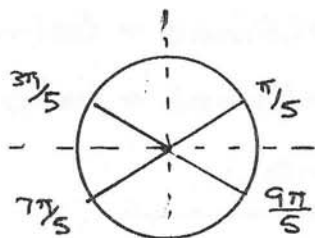
$$= -8i + (-8) + 8i + 8$$

$$= 0$$

$$= \text{R.H.S}$$

Q10) $z^5 + 1 = 0$
 $z^5 = -1$

roots are equally spaced around the unit circle starting at $\frac{\pi}{5}$



(3)

$$z_1 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$z_2 = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

$$z_3 = \cos \pi + i \sin \pi = -1$$

$$z_4 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}$$

$$z_5 = \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}$$

Ques 11 a)

$$(\cos \theta + i \sin \theta)^4 = (\cos \theta + i \sin \theta)^4$$

$$\therefore \cos 4\theta + i \sin 4\theta$$

$$= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

$$= \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$$

$$+ i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

$\therefore$ equating real parts

$$\cos 4\theta = \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$$

b) $\cos 4\theta = \cos^4 \theta + \sin^4 \theta - 6 \cos^2 \theta \sin^2 \theta$

$$= \cos^4 \theta + (1 - \cos^2 \theta)^2 - 6 \cos^2 \theta (1 - \cos^2 \theta)$$

$$= \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

c) $8 \cos^4 x - 8 \cos^2 x + 1 = 0$

ie using (b)

$$\cos 4x = 0$$

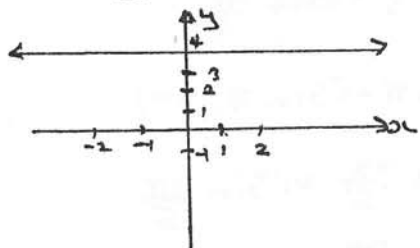
$$4x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \dots$$

$$x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

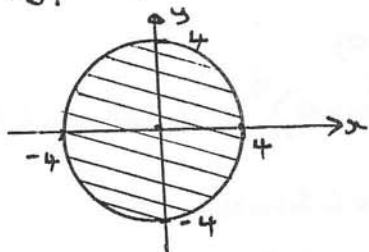
$$\text{for } 0 \leq x \leq \pi$$

Ques 12

a) $\text{Im}(z) = 4$

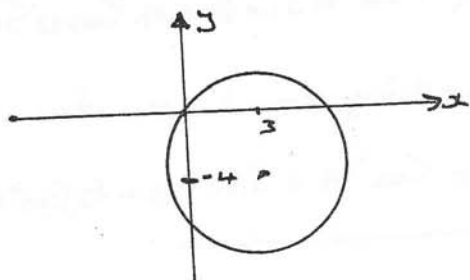


b) $|z| \leq 4$

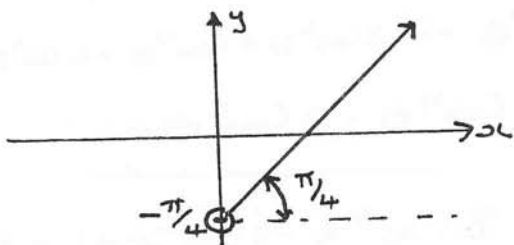


c) $|z - (-3 + 4i)| = 5$

$|z - (-3 + 4i)| = 5$



d) $\arg(z + i) = \frac{\pi}{4}$



e) $z\bar{z} - 4(z + \bar{z}) = 10$

Let $z = x + iy$

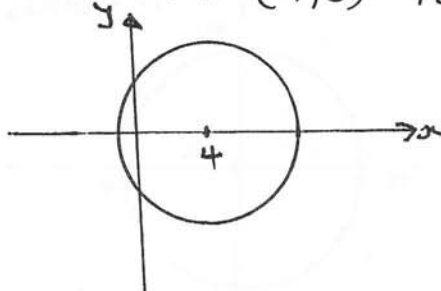
$(x + iy)(x - iy) - 4(x + iy + x - iy) = 10$

$x^2 + y^2 - 8x = 10$

$x^2 - 8x + 16 + y^2 = 10 + 16$

(4)

$(x - 4)^2 + y^2 = 26$

Circle centre $(4, 0)$ radius $\sqrt{26}$ Ques 13

$z^2 + (1 + i)z + k = 0$

 $(1 - 2i)$ is a root

$(1 - 2i)^2 + (1 + i)(1 - 2i) + k = 0$

$1 - 4i - 4 + 1 - 2i + i + 2 + k = 0$

$k = 5i$

Let the other root be β

$\therefore \beta + (1 - 2i) = -\frac{b}{a}$

$\beta + 1 - 2i = -1 - i$

$\beta = -2 + i$

Q14) $z = \cos\theta + i\sin\theta$

$z^n + \frac{1}{z^n} = 2\cos n\theta$

L.H.S

$= z^n + z^{-n}$

$= (\cos\theta + i\sin\theta)^n + (\cos\theta + i\sin\theta)^{-n}$

$= \cos n\theta + i\sin n\theta + \cos(-n\theta) + i\sin(-n\theta)$

$= \cos n\theta + i\sin n\theta + \cos n\theta - i\sin n\theta$

$= 2\cos n\theta$

$$b) \left(z + \frac{1}{z}\right)^3$$

$$= z^3 + 3z^2 \cdot \frac{1}{z} + 3z \cdot \frac{1}{z^2} + \frac{1}{z^3}$$

$$= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$$

$$= z^3 + \frac{1}{z^3} + 3\left(z + \frac{1}{z}\right)$$

$$= 2\cos 3\theta + 6\cos \theta$$

$$\text{also } \left(z + \frac{1}{z}\right)^3$$

$$= (2\cos \theta)^3$$

$$= 8\cos^3 \theta$$

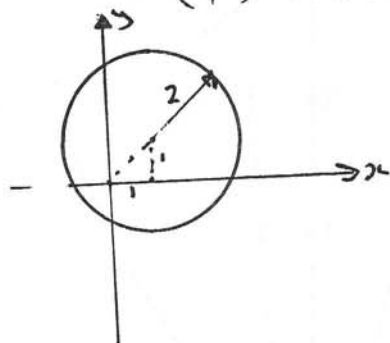
$$8\cos^3 \theta = 2\cos 3\theta + 6\cos \theta$$

$$\cos^3 \theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos \theta.$$

$$15) |z - 1 - i| \leq 2.$$

$$|z - (1+i)| \leq 2.$$

Circle centre $(1,1)$ radius 2.



from the diagram

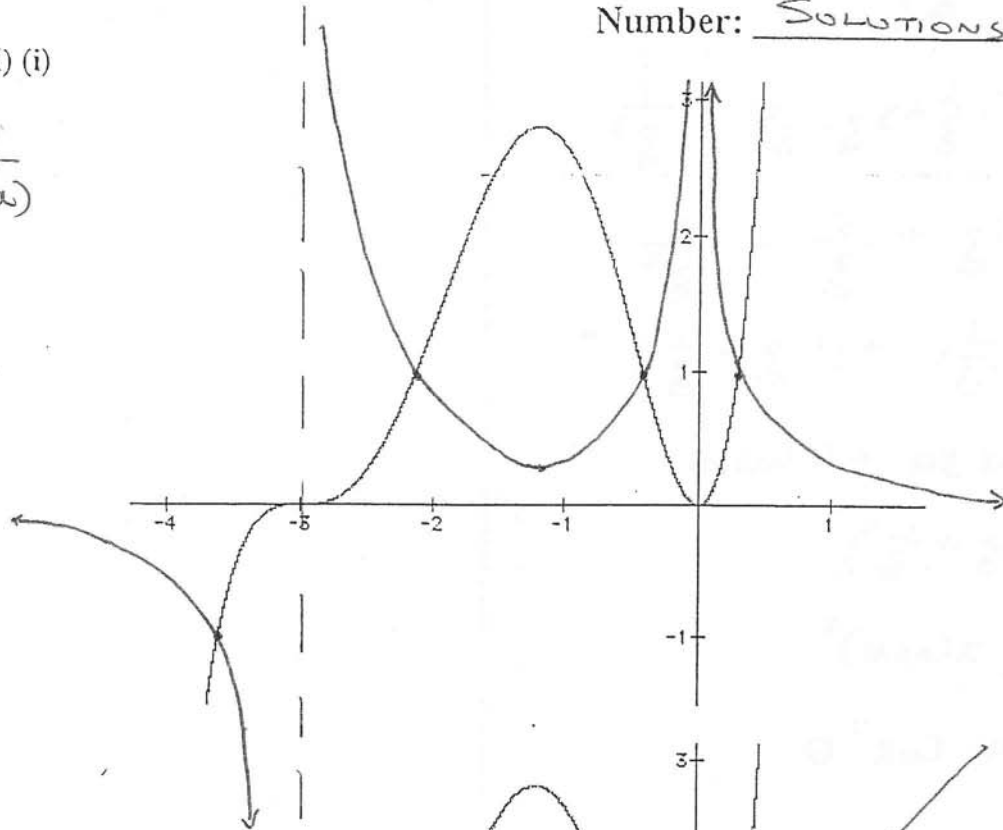
$$\text{max value } |z| = \sqrt{2} + 2.$$

PART B

Number: SOLUTIONS

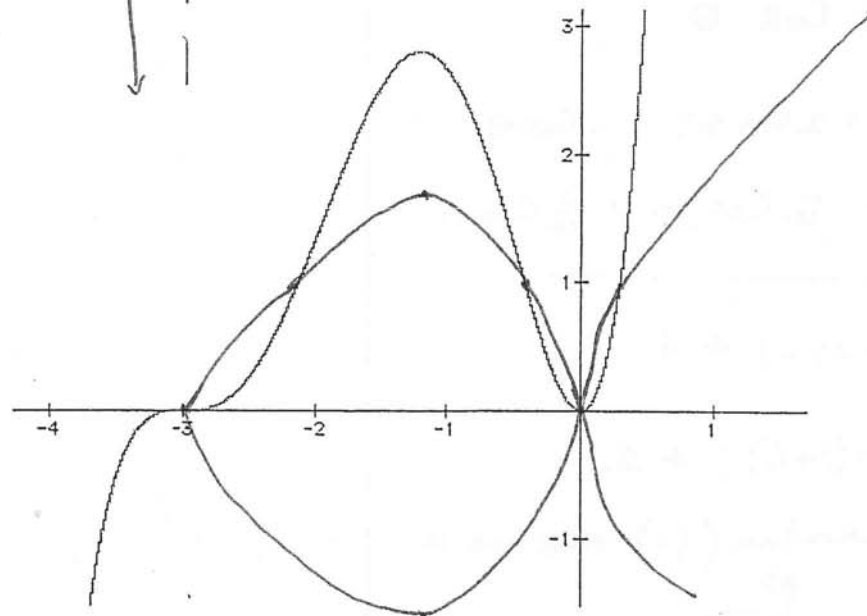
Question 1. (a) (i)

$$y = \frac{1}{f(x)}$$



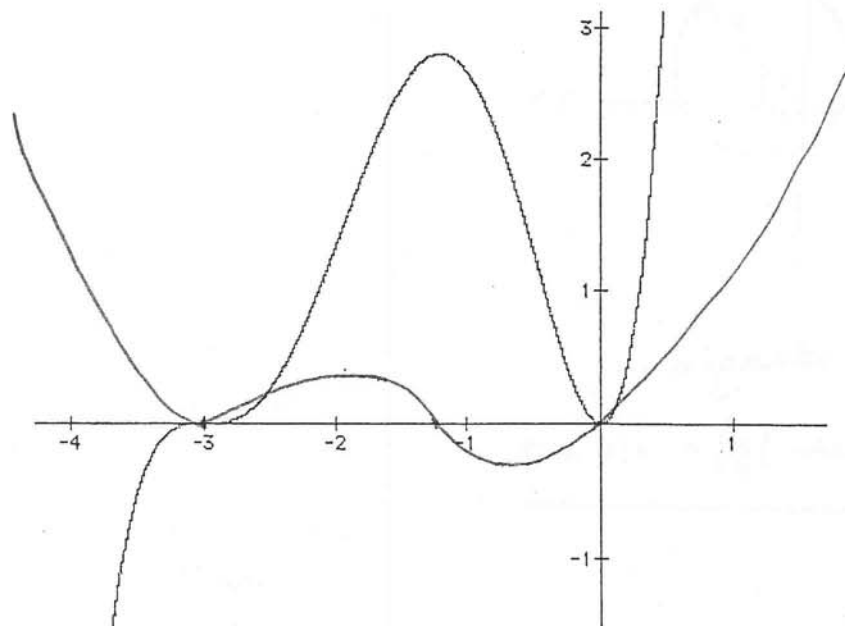
(ii)

$$y^2 = f(x)$$



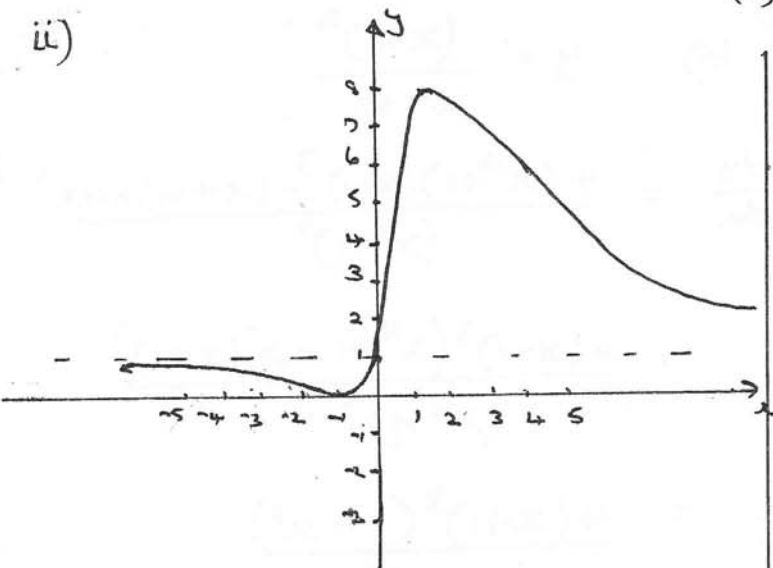
(iii)

$$y = f'(x)$$



(b) $y = x^2(x+3)^3$

ii)



iii) $0 < k < 1$, $1 < k < 8$

Question 3

$$x = \frac{3t}{1+t^3} \quad y = \frac{3t^2}{1+t^3}$$

$$(1+t^3)x = 3t \quad \dots (1)$$

$$(1+t^3)y = 3t^2 \quad \dots (2)$$

$$(2) \div (1) \quad \frac{y}{x} = t$$

Sub into (1)

$$\left(1 + \frac{y^3}{x^3}\right)x = \frac{3y}{x}$$

$$x + \frac{y^3}{x^2} = \frac{3y}{x}$$

$$x^3 + y^3 = 3xy$$

The curve is Symmetrical about $y=x$ because when you interchange x & y you have the same equation.

(7)

ii) $x^3 + y^3 = 3xy$

$$3x^2 + 3y^2 \frac{dy}{dx} = 3\left(x \cdot \frac{dy}{dx} + y\right)$$

$$3x^2 + 3y^2 \frac{dy}{dx} = 3x \frac{dy}{dx} + 3y$$

$$\frac{dy}{dx} (3y^2 - 3x) = 3y - 3x^2$$

$$\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x}$$

$$= \frac{y - x^2}{y^2 - x}$$

horizontal tangent when

$$\frac{dy}{dx} = 0$$

$$\frac{y - x^2}{y^2 - x} = 0$$

$$y - x^2 = 0$$

$$y = x^2$$

$$\therefore x^3 + (x^2)^3 = 3x(x^2)$$

$$x^3 + x^6 = 3x^3$$

$$x^6 - 2x^3 = 0$$

$$x^3(x^3 - 2) = 0$$

$$x = 0, \quad x = \sqrt[3]{2}$$

Vertical tangent $\frac{dx}{dy} = 0$

$$\frac{y^2 - x}{y - x^2} = 0$$

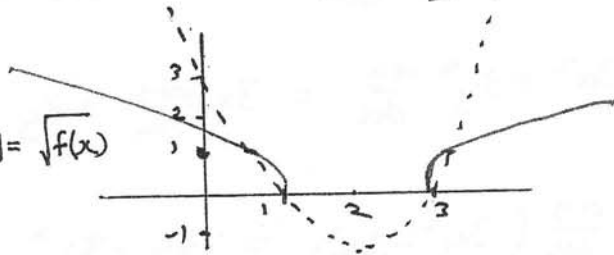
$$\text{i.e. } y^2 = x$$

Question 2.

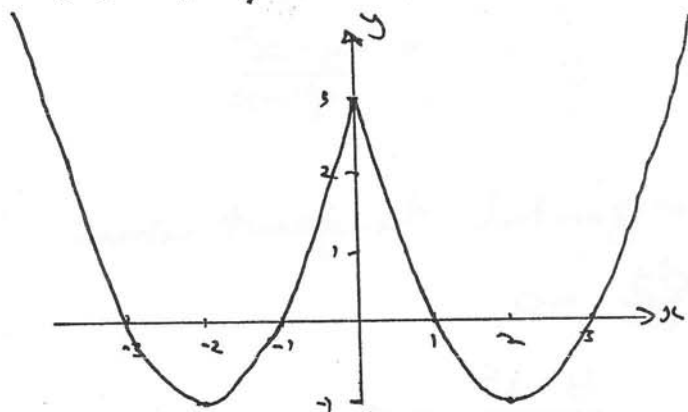
PART B

a) $f(x) = (x-1)(x-3)$

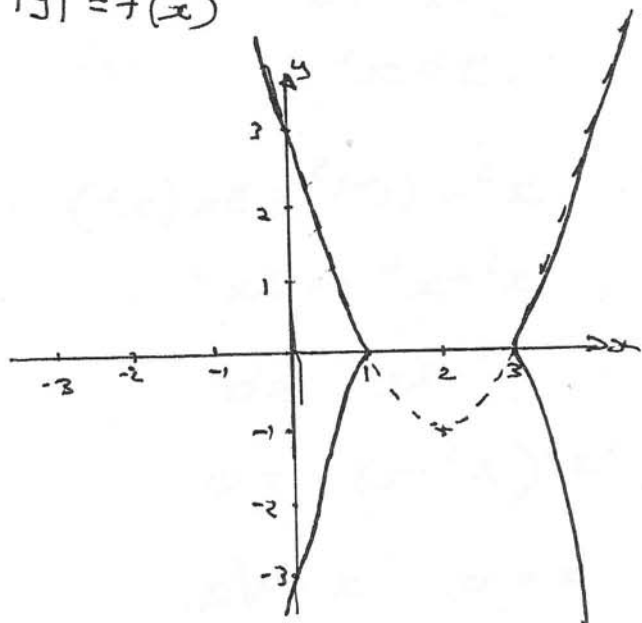
i) $y = \sqrt{f(x)}$



ii) $y = f(|x|)$



iii) $|y| = f(x)$



(b)

b) $y = \frac{(x+1)^4}{x^4+1}$

$$\frac{dy}{dx} = \frac{4(x^4+1)(x+1)^3 - (x+1)^4 \cdot 4x^3}{(x^4+1)^2}$$

$$= \frac{4(x+1)^3(x^4+1 - x^3(x+1))}{(x^4+1)^2}$$

$$= \frac{4(x+1)^3(1-x^3)}{(x^4+1)^2}$$

S.P. $\frac{dy}{dx} = 0$

$$4(x+1)^3(1-x^3) = 0$$

$$\therefore x = -1, 1$$

$$f'(-2) < 0$$

$$f'(2) < 0$$

$$f'(0) > 0$$

$$f'(0) > 0$$

$$\therefore (-1, 0)$$

min

$$\therefore (1, 8)$$

max

asymptotes

$$\frac{(x+1)^4}{x^4+1}$$

$$= \frac{x^4 + 4x^3 + 6x^2 + 4x + 1}{x^4 + 1}$$

$$= \frac{1 + \frac{4}{x} + \frac{6}{x^2} + \frac{4}{x^3} + \frac{1}{x^4}}{1 + \frac{1}{x^4}}$$

$$\text{as } x \rightarrow \infty \quad y \rightarrow 1.$$

$$\therefore \text{asymptote } y = 1.$$

$$\therefore y^6 + y^3 = 3y^3$$

$$y^6 - 2y^3 = 0$$

$$y^3(y^3 - 2) = 0$$

$$y = 0, \sqrt[3]{2}.$$

$$\text{now } y^2 = x$$

$$\therefore x = (\sqrt[3]{2})^2$$

$$= \sqrt[3]{4}$$

$\therefore$ vertical tangent $(\sqrt[3]{4}, \sqrt[3]{2})$